Manoj Sai Penugonda

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SUMMARY

Data Scientist with 3.8 years of experience building and deploying scalable Machine Learning and Generative AI solutions. Strong expertise in data processing, feature engineering, predictive modeling, applied GenAI use cases, prompt engineering, transforming raw data into actionable insights that improve business outcomes and collaborating effectively with stakeholders for strategic decision-making.

EDUCATION

Anna University 08/2018 - 07/2022
Madras Institute of Technology *B.E. in Electronics Engineering*

EXPERIENCE

Data Scientist II 06/2026 – Present
Ericsson

- Owned the end-to-end development of an AI/ML platform for proactive telecom network performance management across Ericsson's **Radio Access Network (RAN)**, processing large-scale KPI data from **thousands of cell sites**.
- Built a **LightGBM time-series regression model** to predict throughput degradation **up to 2 hours in advance**, enabling proactive remediation before subscriber service impact.
- Designed an automated **Root Cause Analysis (RCA)** framework combining **Isolation Forest**, **Quantile Regression**, **PCA**, and **LightGBM** to detect KPI anomalies, isolate fault-causing network modules, and generate corrective parameter recommendations.
- Automated anomaly detection and recommendation workflows, reducing manual troubleshooting effort by **~60%** and significantly improving Mean Time to Resolution (MTTR).
- Deployed production-grade ML pipelines on **Google Cloud Platform** using **Docker**, **Kubernetes**, and **Argo Workflows**, delivering scalable, fault-tolerant, and reproducible model execution
- Collaborated with cross-functional data science, platform engineering, and telecom domain teams to integrate predictive analytics into Ericsson's proactive network optimization ecosystem.

Data Scientist 10/2022 – 06/2026
Accenture *Chennai*

- Built **regression-based ML models** that achieved **85% accuracy** in predicting weekly product sales volume based on promotional data and historical trends.
- Spearheaded **classification-based ML models** achieving an **88% AUC-ROC** score to predict patient readmission risk by analyzing complex clinical records, enabling early intervention for high-risk patients.
- Developed a **Generative AI-**based chatbot utilizing a **Retrieval-Augmented Generation (RAG)** framework, integrating **Hugging Face embeddings (thenlper/gte-small)** and **ChromaDB** to improve enterprise search accuracy by **25%** over traditional keyword methods.
- Optimized vector search latency by **20%** through advanced text chunking strategies and refined similarity search parameters, enhancing retrieval speed and relevance for large-scale datasets.
- Designed a user-friendly GUI using **Streamlit**, reducing decision-making time for non-technical stakeholders by **40%** and improving accessibility.

TECHNICAL SKILLS

Programming and Query Languages: Python, SQL
Machine Learning & Gen AI: Linear & Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, AdaBoost, Neural Networks (ANN, CNN, RNN), Transformers, Generative AI, LLM Fine-Tuning (LoRA), GANs, Diffusion Models, NLP, Clustering, Optimization Techniques
Data Science Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow
Tools & Technologies: Docker, Kubernetes, Git, GitHub, RESTful APIs, ReactJS, FastAPI, Google Cloud Platform (GCP)
Analytics: Statistical Analysis, Data Visualization, Data Wrangling, Feature Engineering, A/B Testing, Hypothesis Testing

CERTIFICATIONS

- GH-300 GitHub Copilot.